



Understanding Gene Behavior Across Cancer Types

Alex Nite

MSA 530

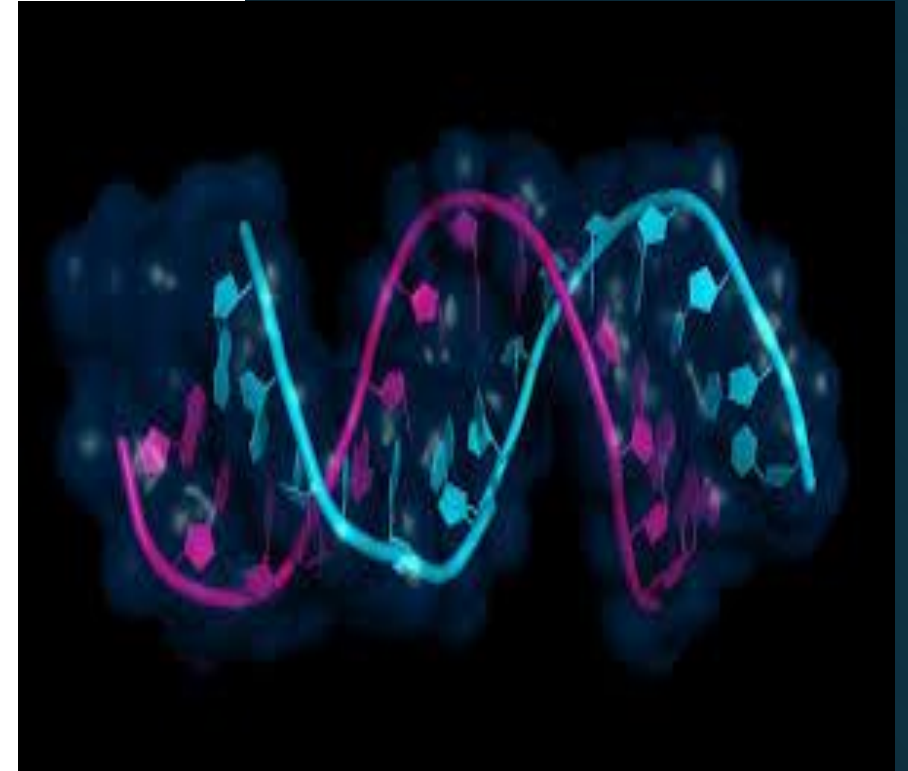
Problem / Question

Goal

Identify patterns in gene expression, variability, and network connectivity across cancers.

Key questions

- Which genes are consistently highly expressed?
- Which genes are highly variable?
- Which genes are central in gene networks?

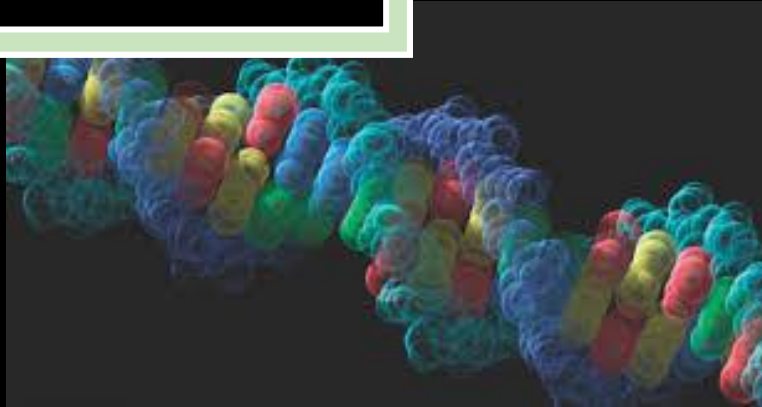
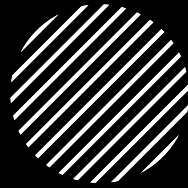


Why This Matters?

- Gene expression informs cell behavior and cancer progression.
- Consistently active genes may be essential for survival or tumor growth.
- Variable genes may respond to stress or the environment.
- Central genes can control key pathways.



Data Source

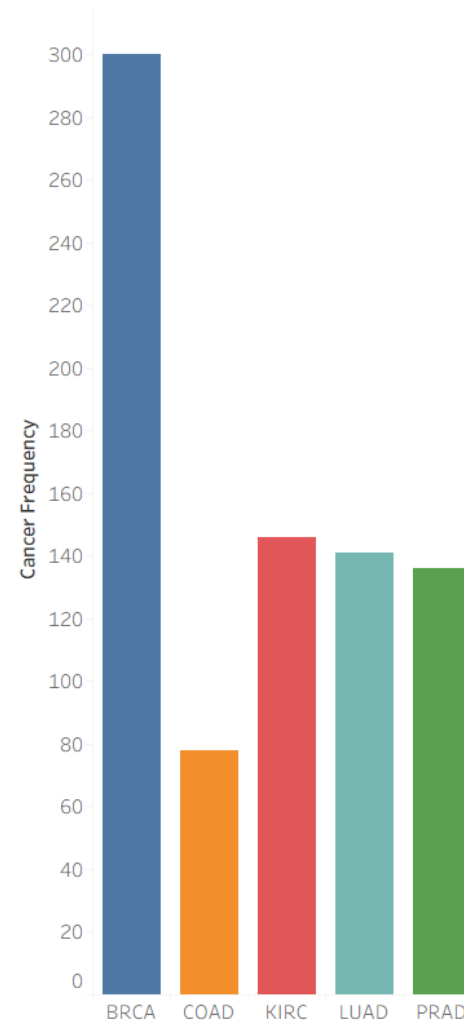


- Original dataset: Cancer Genome Atlas Research Network
 - Project : Cancer Genome Atlas Pan-Cancer et al., 2013
- Accessed via Nyantudre (2025) on Kaggle
- 5 Cancer types
 - BRCA - Breast Invasive Carcinoma
 - COAD - Colon Adenocarcinoma
 - KIRC - Kidney Renal Clear Cell Carcinoma
 - LUAD - Lung Adenocarcinoma
 - PRAD - Prostate Adenocarcinoma (PRAD).
- 15 Randomly Chosen Genes
 - G1, G2, G3, G59, G60, G61, G2359, G2360, G2361, G2362, G2363, G2364, G2421, G2452, G3427
- Dataset has a total of 801 Data points

Sample Distribution by Cancer

Cancer Type Distribution

Frequency of
Cancer Type



Class (Gene Data)

- ☒ (All)
- ☒ BRCA
- ☒ COAD
- ☒ KIRC
- ☒ LUAD
- ☒ PRAD

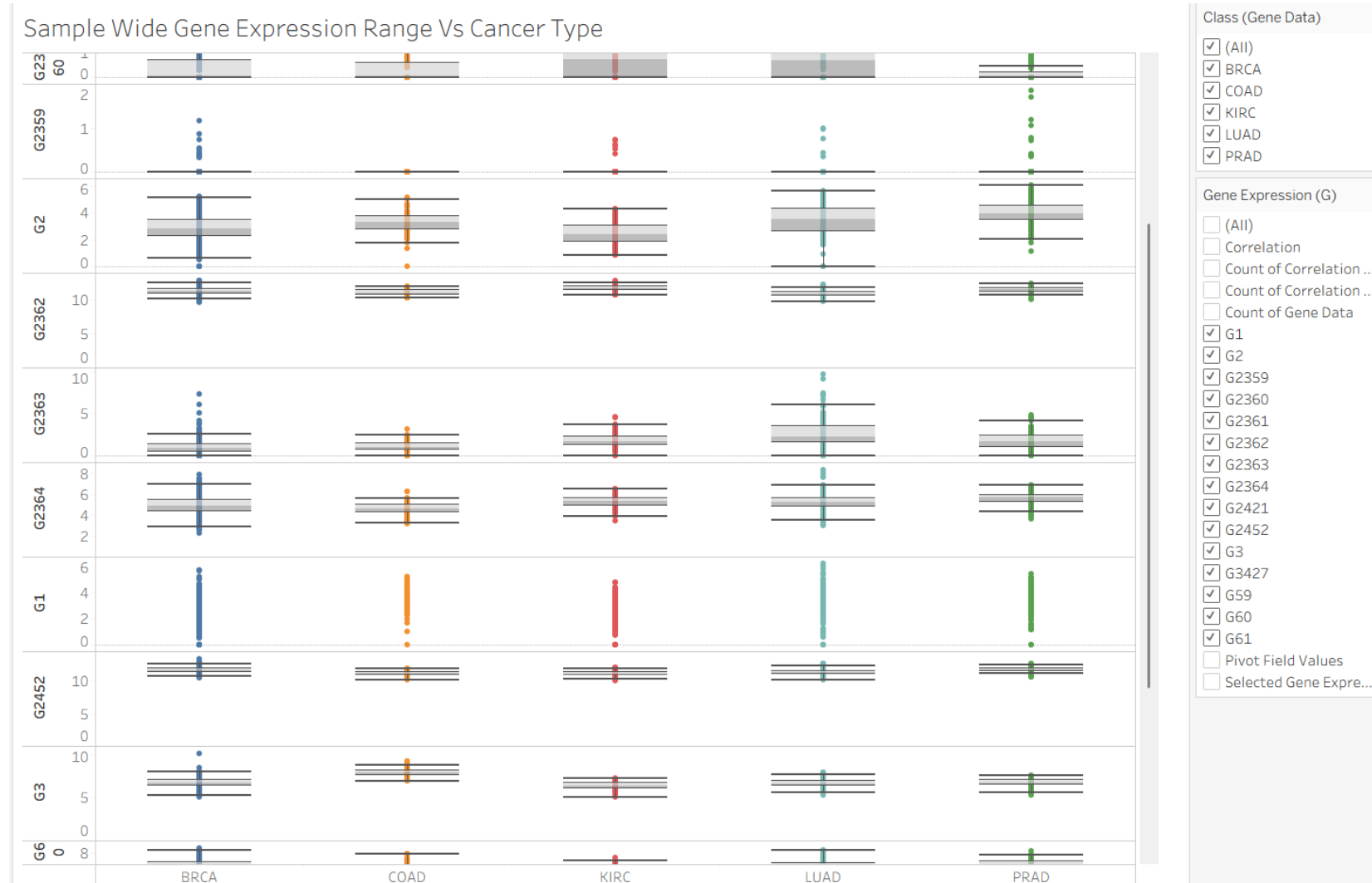
Class (Gene Data)

- BRCA
- COAD
- KIRC
- LUAD
- PRAD

BRCA is the most frequent
cancer type in this sample
while COAD is the least

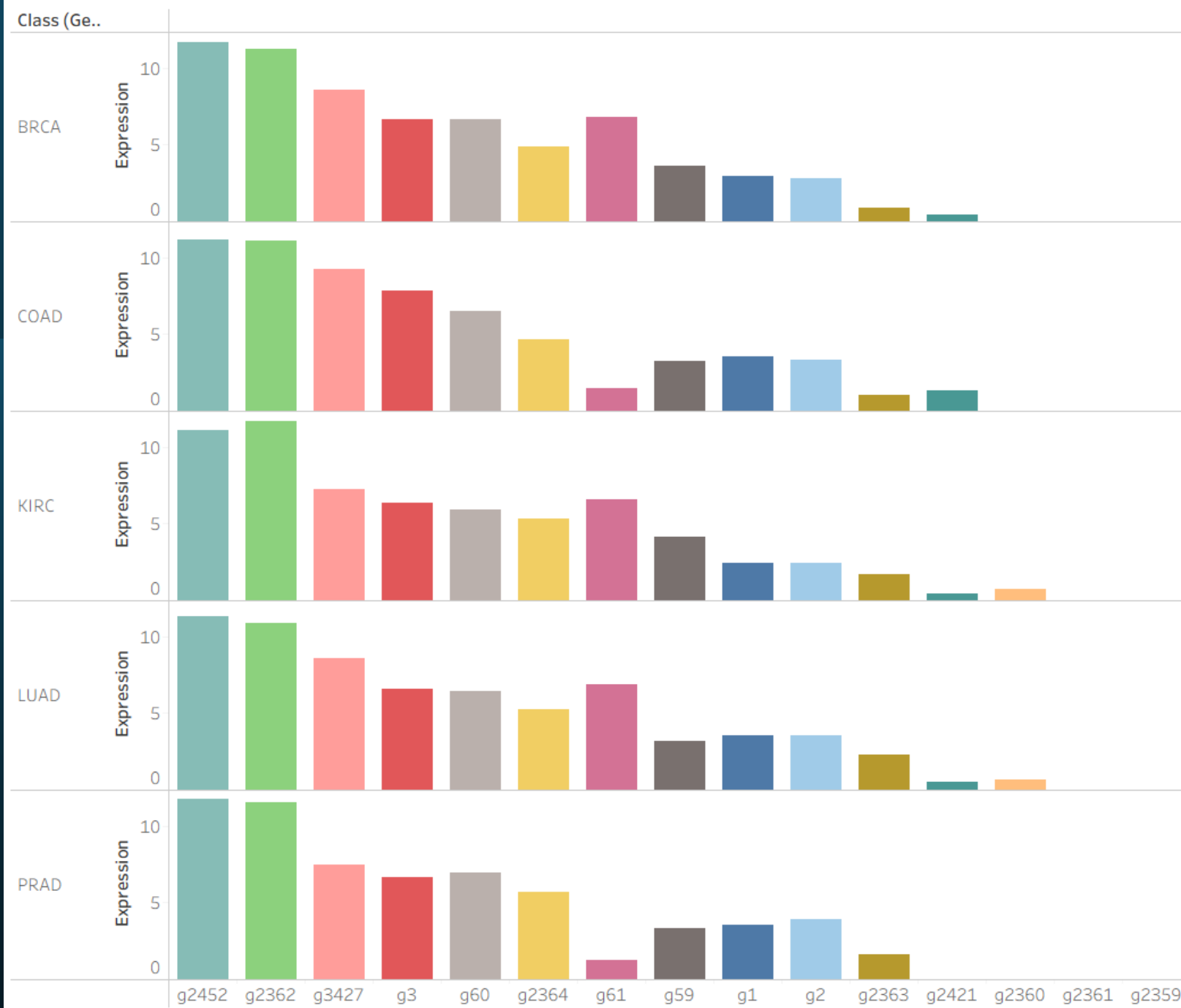
Gene Expression Range by Cancer Type across sample

- Examining genes by cancer type, G2452 was highly expressed in most cancers except KIRC, where G2362 slightly led.
- G61 remained the most variable across all cancers.
- PRAD had low expression and low variability, while LUAD and BRCA showed higher expression and variability.

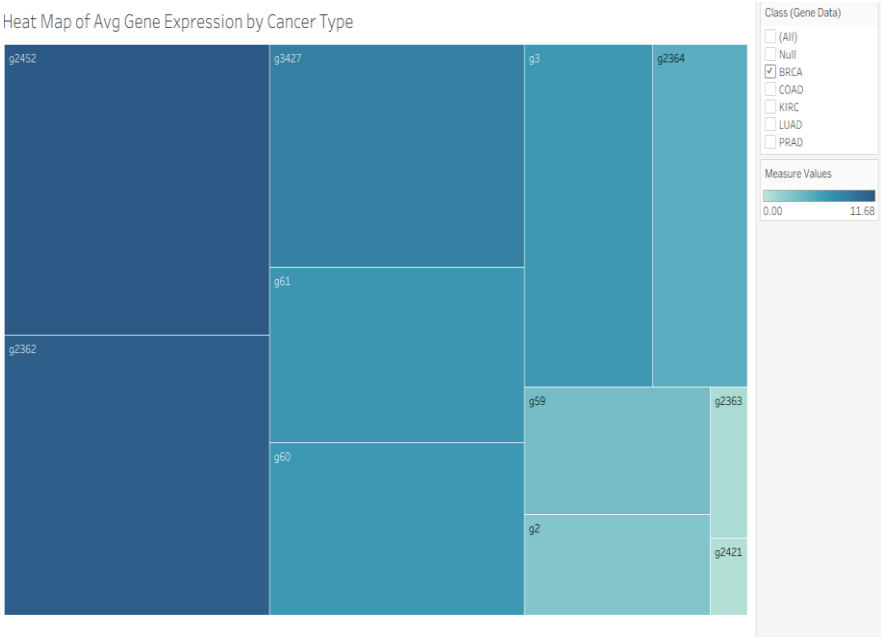


Avg Gene Expression by Cancer Type

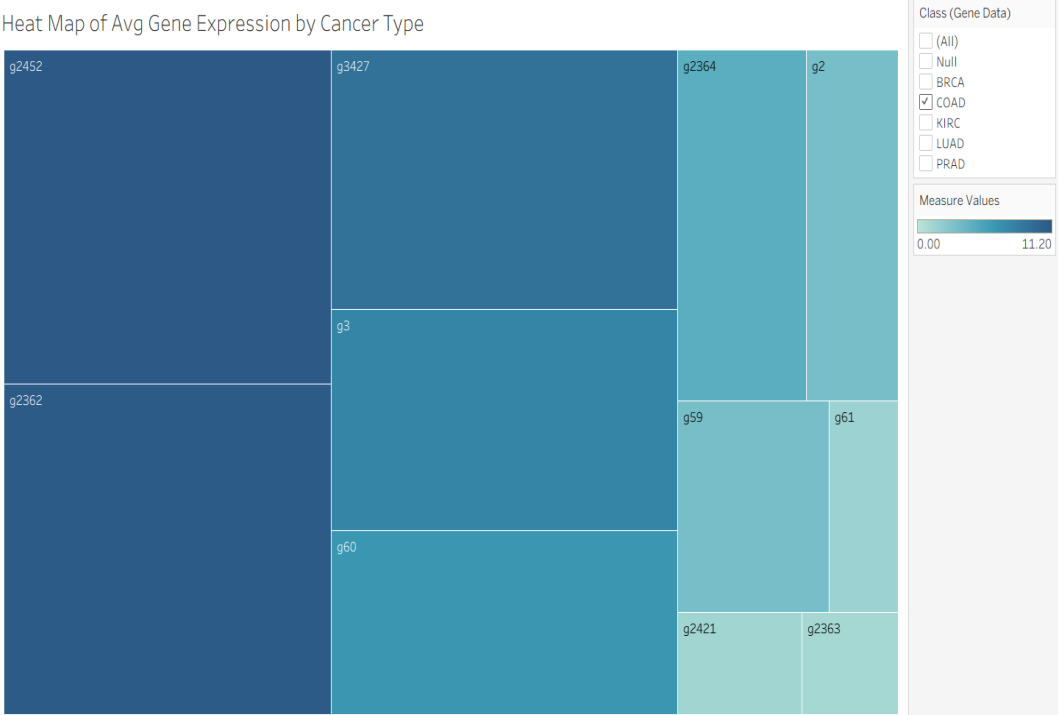
Average Gene Expression by Cancer Type



Heat MAP: Average BRCA and COAD Gene Expression

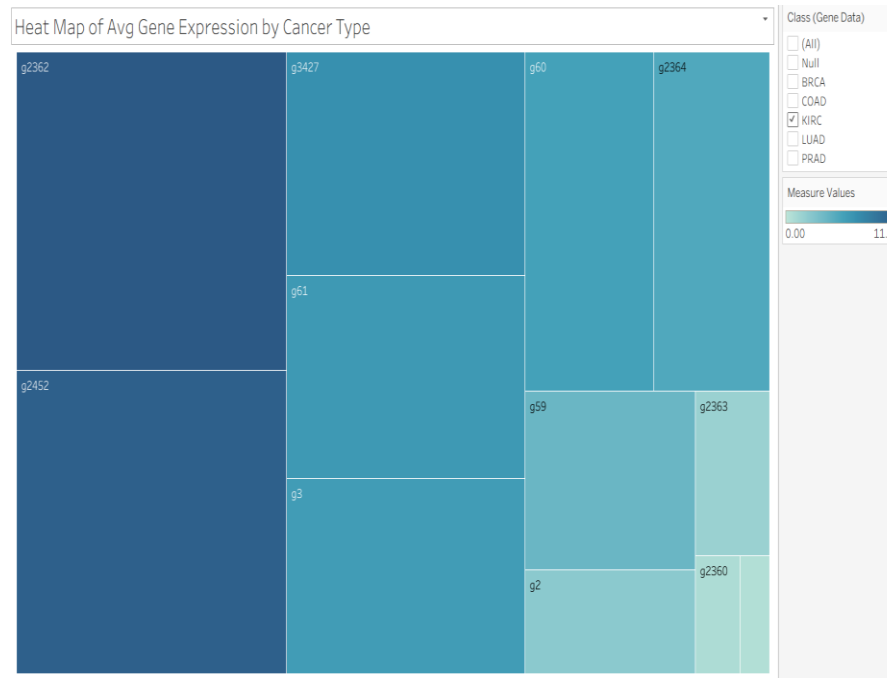


BRCA



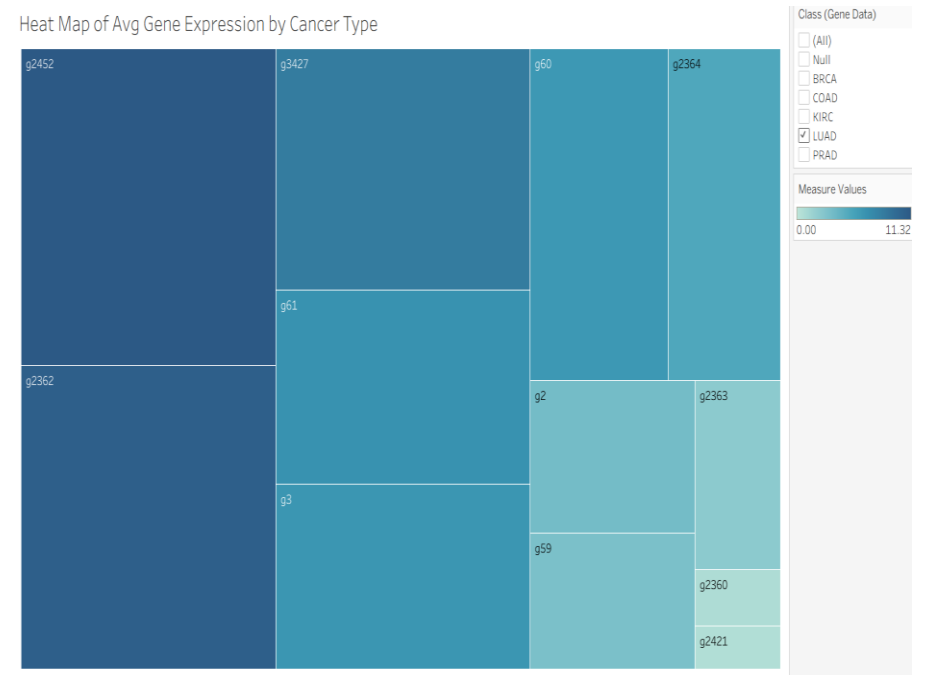
COAD

Heat MAP: Average KRIC and LAUD Gene Expression



KRIC

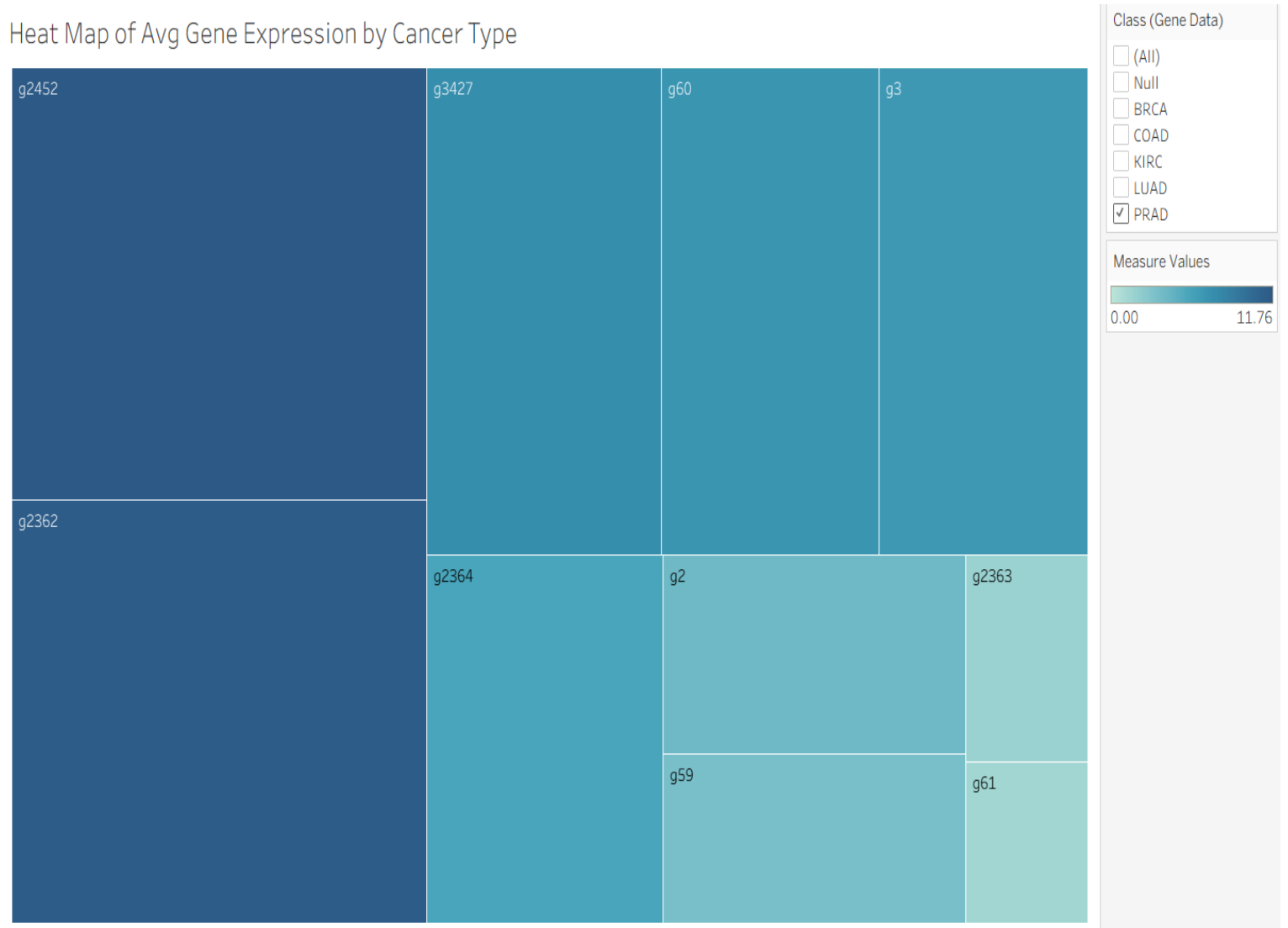
Heat Map of Avg Gene Expression by Cancer Type



LAUD

Heat MAP: Average PRAD Gene Expression

Heat Map of Avg Gene Expression by Cancer Type



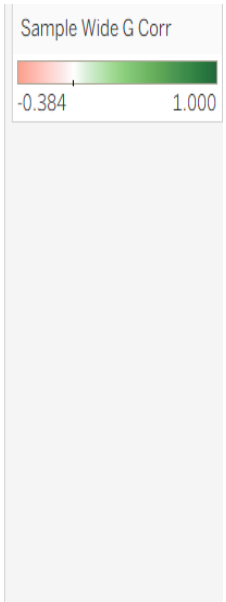
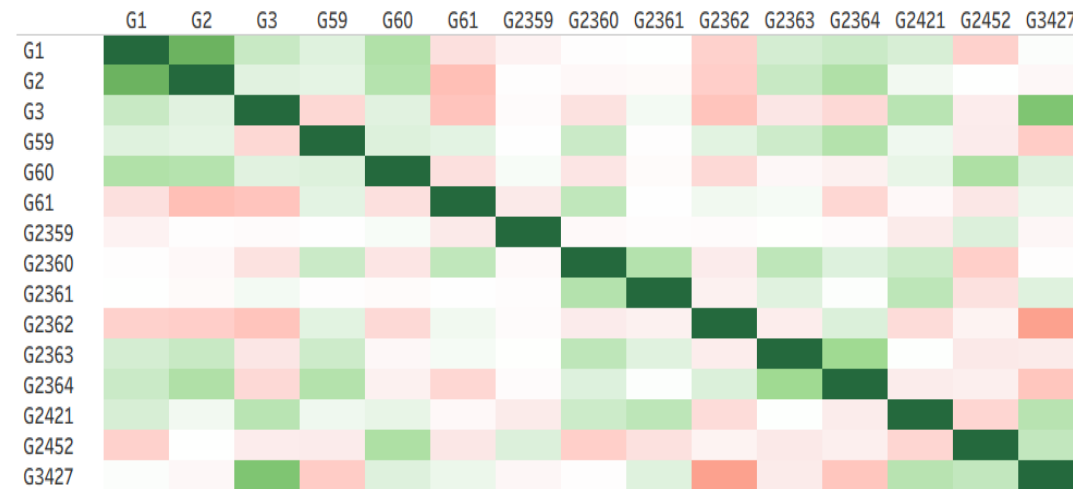
Insights of Average Gene Expression

- G2452 was the most highly expressed in BRCA, COAD, LUAD, and PRAD.
 - KIRC was a little different, with G2362 being slightly more expressed than G2452.
- G2359 is silent in all types of cancer but KIRC.
- G61 was the most variable in all cancers, responding strongly to differences in samples.
- In general, cancers with higher expressions like LUAD and BRCA also had higher variability, while PRAD had low expression and low variability.



Sample Wide Gene Correlation

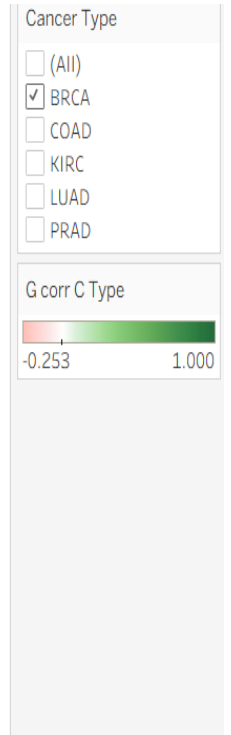
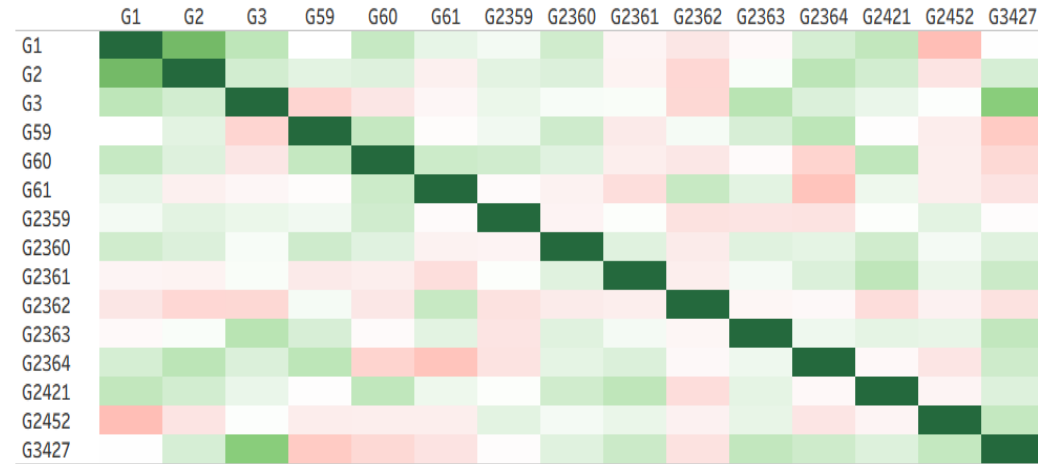
Gene Correlation Matrix for overall dataset



- G1 and G2 were strongly positively correlated, suggesting that across the sample they work together.
- Some genes showed strong negative correlations, like G2362 with G3427 indicating that they don't express together.
- G3 appeared central in the network, while G2359 remained independent.

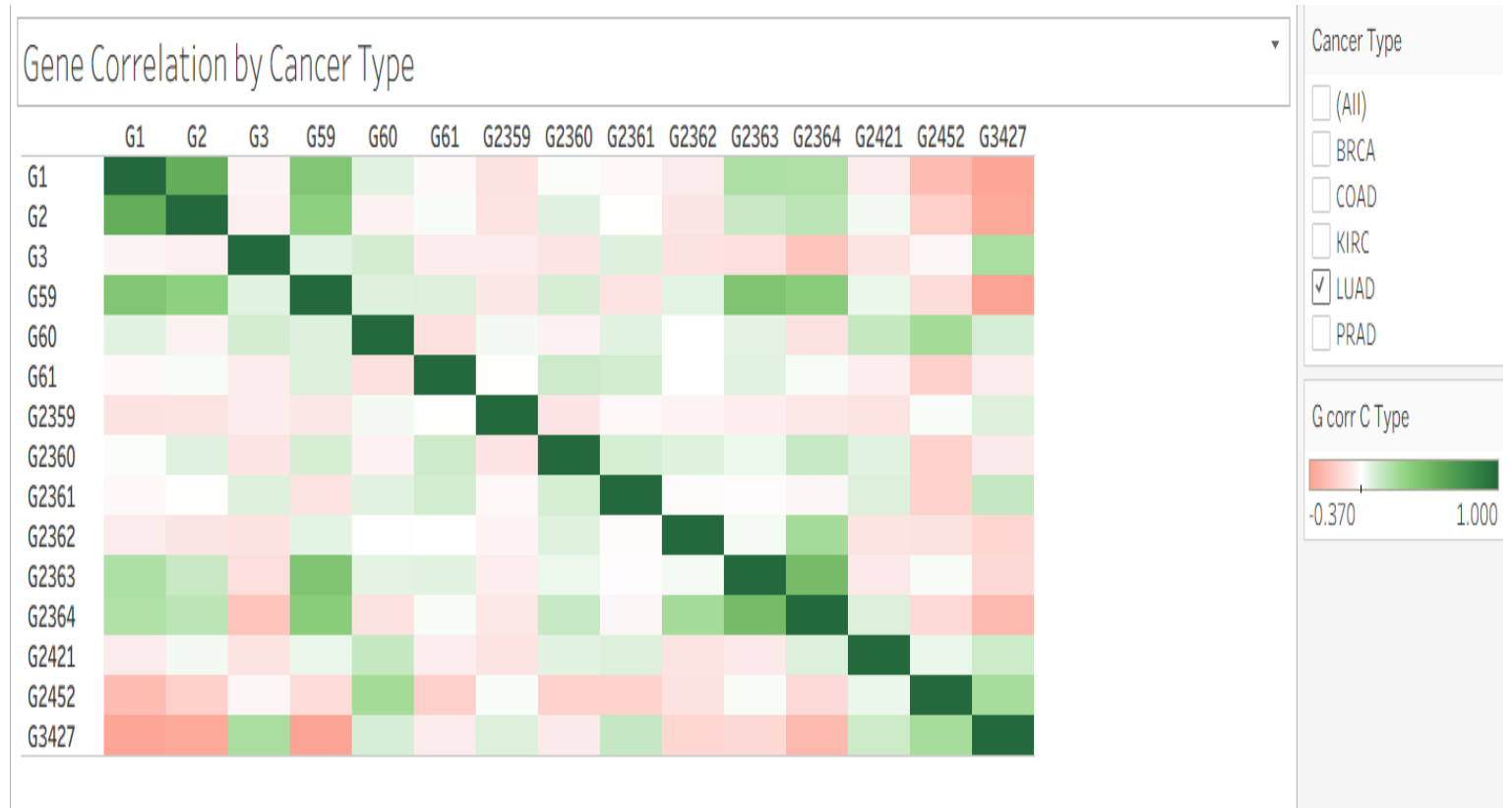
Gene Correlation by Cancer Type : BRCA

Gene Correlation by Cancer Type



- In BRCA, G1 and G2 formed the main co-expression pair, showing a strong positive correlation ($R=0.488$).
- For this cancer, the negative links between genes included, G1 with G2452 ($R=0.253$).
- G3427 was the most central gene in the network, while G2359 was largely independent.

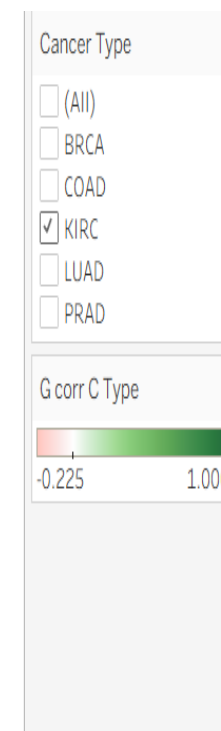
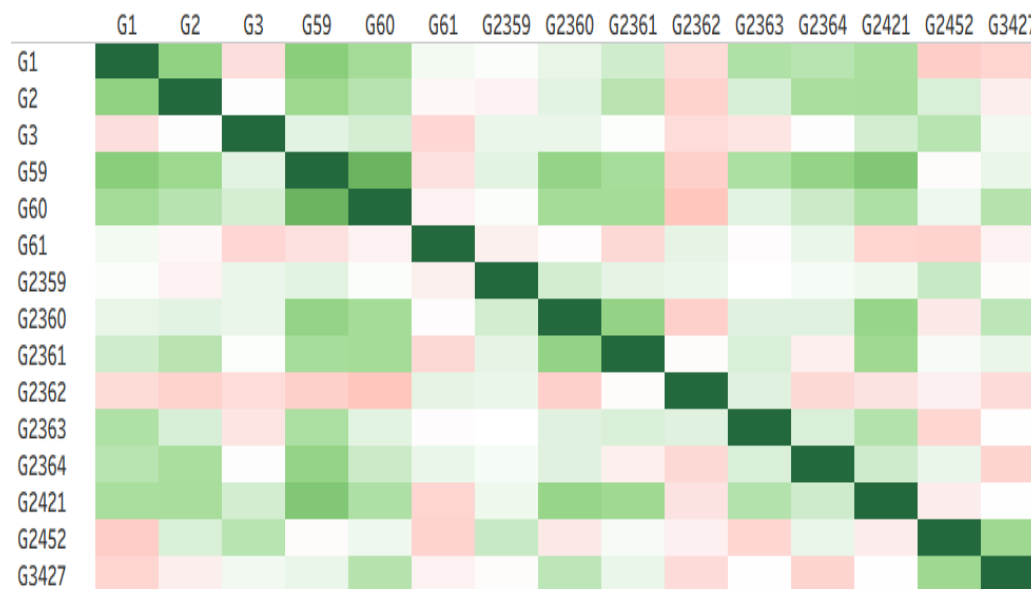
Gene Correlation by Cancer Type : LUAD



- In LUAD, G1 and G2 had the strongest positive correlation ($R = 0.578$).
- G2363 and G2364 also formed a tight co-expression cluster ($R=0.482$).
- For this type of cancer, G3427 drove the strongest negative correlations and often acted as a regulatory sink. It was the most central gene, while lowly expressed or highly variable genes like G2359 and G61 were the least connected.

Gene Correlation by Cancer Type : KIRC

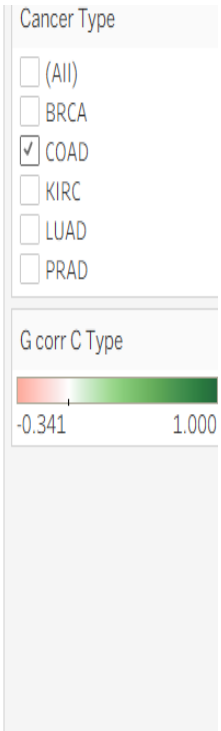
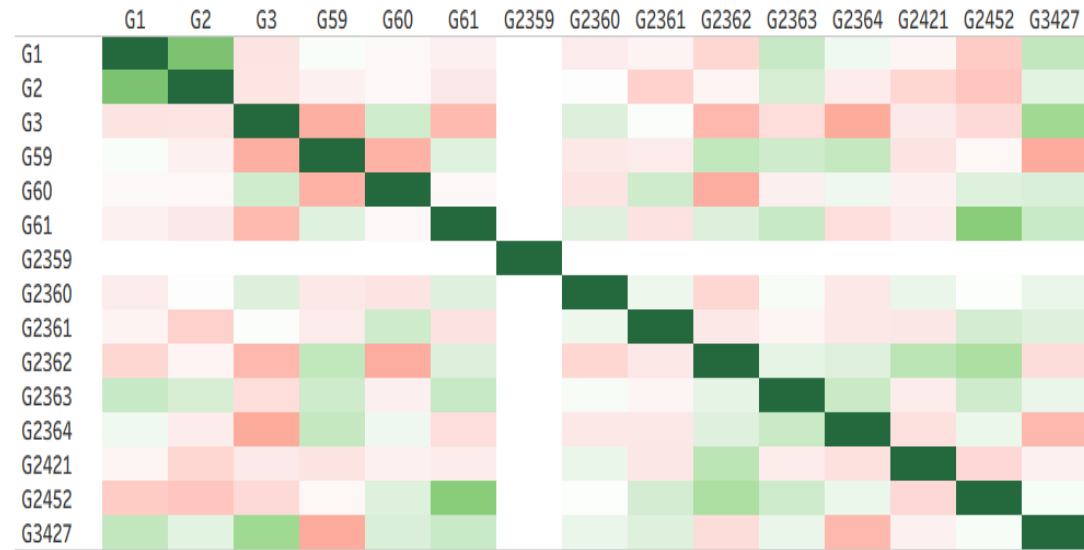
Gene Correlation by Cancer Type



- G59, G60, G1, and G2 form a core co-expression module.
- The strongest positive correlation was between G59 and G60 (R= 0.532).
- G2362 showed negative correlations with this module while G59 was the most central gene in KIRC

Gene Correlation by Cancer Type : COAD

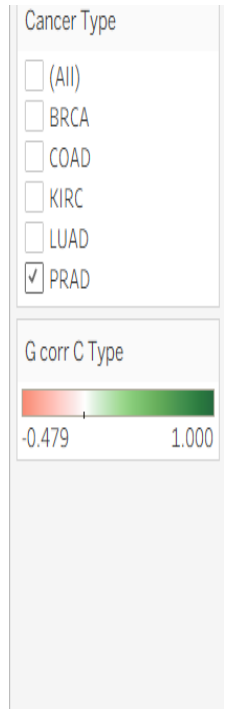
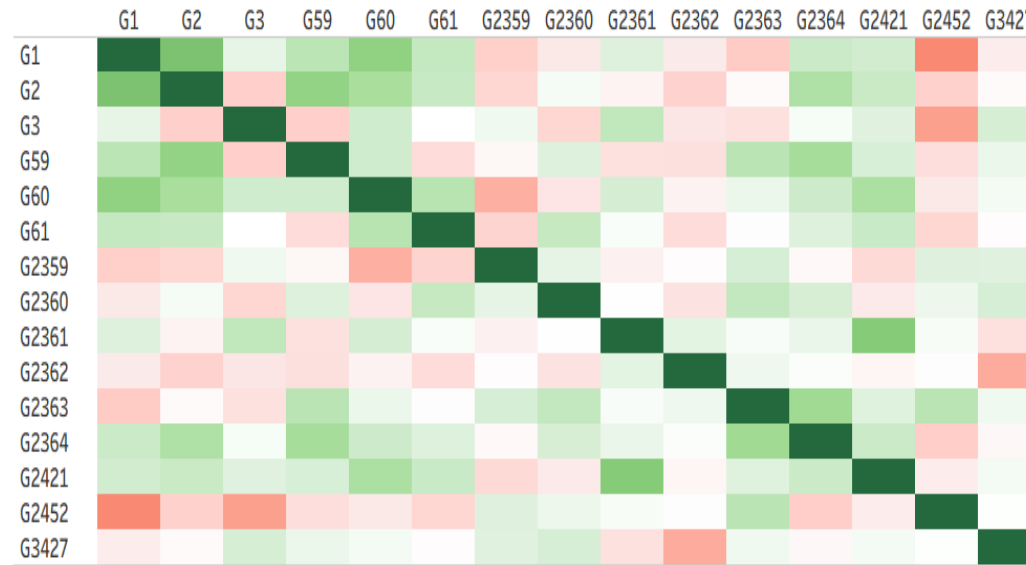
Gene Correlation by Cancer Type



- The strongest positive correlation was between G2452 and G61 ($R = 0.369$).
- The strongest negative correlation was between G2364 and G3 ($R = -0.341$).
- G3 was the most connected gene, while G1 was the least connected.

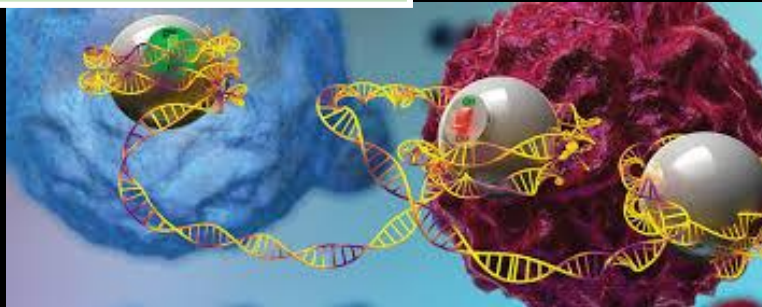
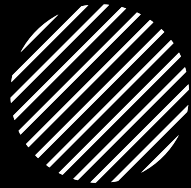
Gene Correlation by Cancer Type : PRAD

Gene Correlation by Cancer Type

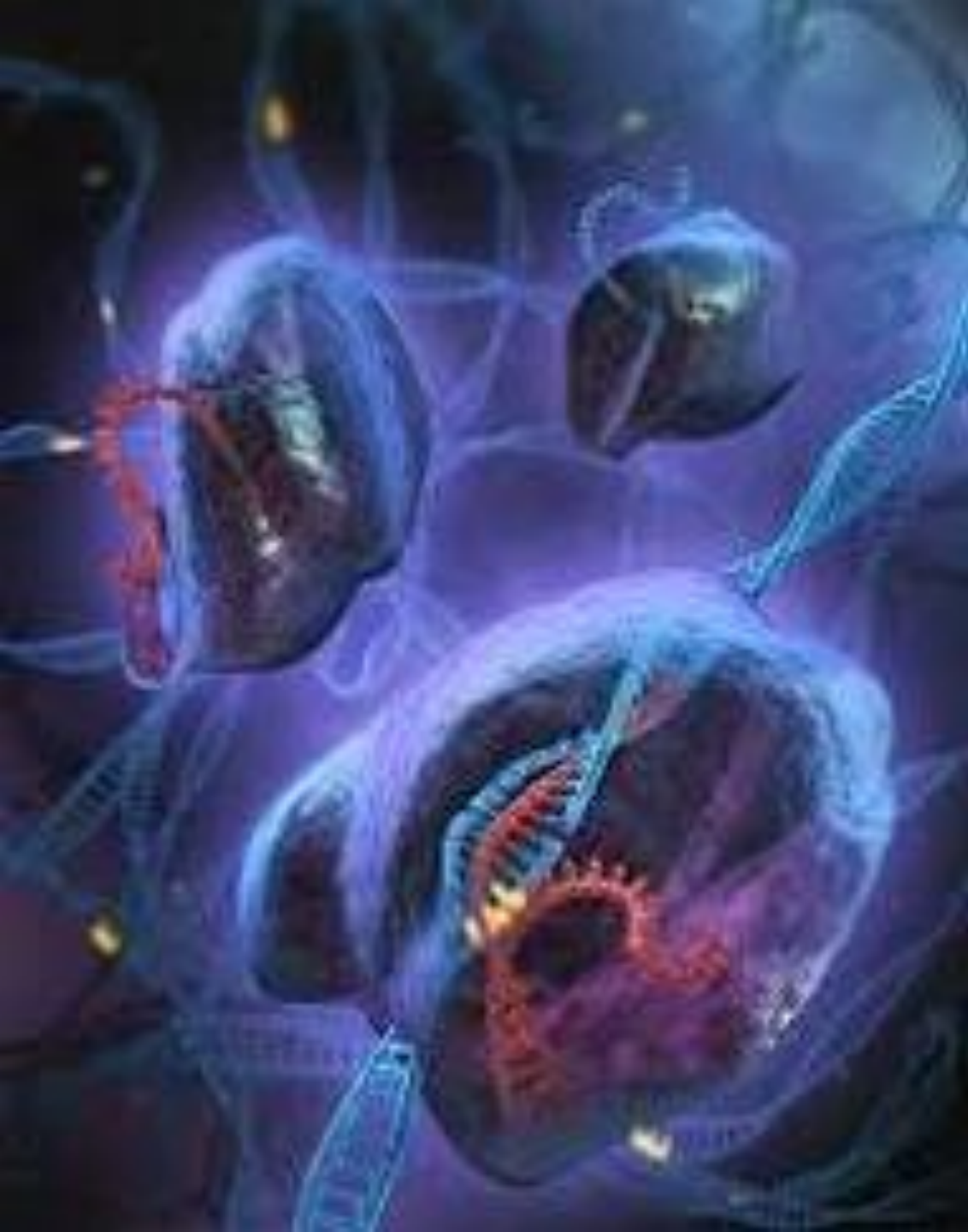


- In **PRAD**, G1 and G2 were positively correlated ($R=0.436$).
- G1 was the most central gene here, while G3427 had limited influence on the network.

Discussion



- Highly expressed: G2452, G2362
- Most variable: G61
- Core modules: G1 and G2
- Independent: G2359
- Antagonistic pairs: G1 & G2452



Challenges

- Small gene subset might miss interactions
- Outlier-heavy genes affect correlations (G2361)
- Tableau correlation required Python pre-processing and pivoting
- Wide-to-long format conversion was needed for correlation analysis for cancer type
- Coordinating multiple Excel sheets

Conclusion



The dataset revealed strong and consistent expression patterns despite analyzing only a small group of genes.

G2452 and G2362 stood out because they were repeatedly highly expressed, which hints at important biological roles in cancer development or progression.

G61 became important for the opposite reason: its high variability tells us it may react differently across patients or tumor environments.

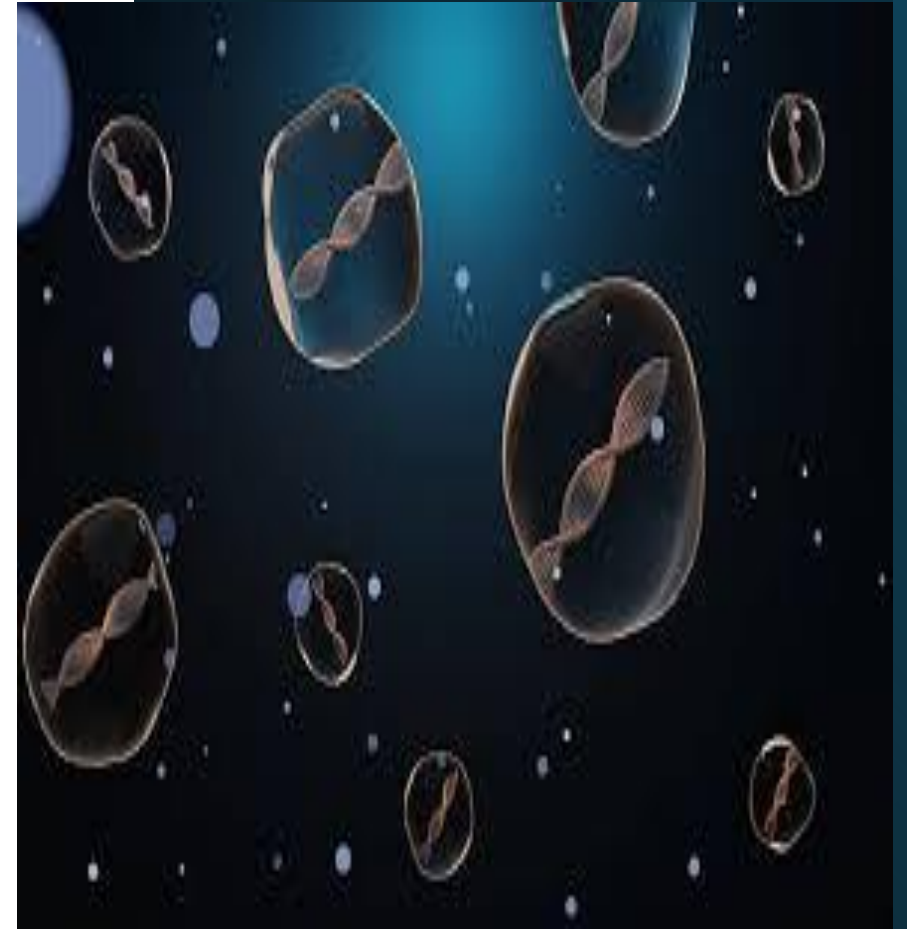
The co-expression findings showed that G1 and G2 consistently travel together across cancers, forming what looks like a shared functional module.

Some genes like G2359 barely connect to others, showing independence.

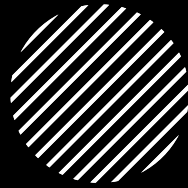
Strong negative relationships, such as G1 vs. G2452, help reveal repression-type patterns or opposing pathways.

Conclusion

- Even with technical challenges, the insights matched what previous literature suggests, meaning the analysis is directionally valid.
- These results help us think about which genes might need more attention in future cancer research or which ones might be meaningful targets for deeper study.



CITATIONS



- Cancer Genome Atlas Research Network, Weinstein, J. N., Collisson, E. A., Mills, G. B., Shaw, K. R., Ozenberger, B. A., Ellrott, K., Shmulevich, I., Sander, C., & Stuart, J. M. (2013). The Cancer Genome Atlas Pan-Cancer analysis project. *Nature Genetics*, 45(10), 1113–1120.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3919969/>
- Nyantudre, A. (2025). *Gene Expression Cancer RNA-Seq [Data set]*. UCI Machine Learning Repository.
<https://www.kaggle.com/datasets/waalbannnyantudre/gene-expression-cancer-rna-seq-donated-on-682016>

Thank You!

